

Per-Particle Spring-Energy Clustering Is Kernel Clustering:

Known Equivalences Behind a Null Screening

Luciano Silva Alarco

Pontificia Universidad Católica del Perú

ORCID 0000-0002-3395-3129

Working draft v0.4 (two rounds of internal review applied to v0.3; the results on invariances and size weights are new and not yet refereed) — research line LMI001 — 3 October 2026

Abstract

A research line on “mechanical analogies for data clustering” screened objectives built from springs, tension and lifting against matched kernel controls and found no surviving mechanical component. No manuscript or code of that screening was available to us: we reconstruct the family from the line’s public summary as *pairwise* spring energies (a pair potential $\phi(d)$, optionally after a lift, normalised per particle or summed) and read the matched control as kernel k -means on the same matrix. For the per-particle normalisation the null result follows from known results, collected here in the language of springs: the energy is the pairwise-clustering cost of Hofmann and Buhmann with dissimilarities $\phi(d_{ij})$, hence, up to a constant and a factor $\frac{1}{2}$, a kernel k -means objective that a constant shift makes positive semidefinite without changing the minimisers (Roth et al.; Dhillon, Guan and Kulis); zero-rest-length Hooke springs give k -means; for potentials of negative type, such as the constant-tension string $\phi(d) = d$, minimising it is kernel k -means with the dataset-independent distance-induced kernel of Sejdinovic et al., i.e. energy-statistics clustering (França, Rizzo and Vogelstein); and zero-temperature relaxation is Hartigan’s method in kernel space (kernel k -groups), whose fixed points are Voronoi-stable, i.e. Lloyd fixed points with ties broken in favour of the current cluster. The summed energy is min-sum k -clustering (max- k -cut), and we show that it is *not* a weighted kernel k -means objective for non-negative springs, even with data-dependent point weights. Beyond the synthesis, our additions are small: the mechanism by which a diagonal shift freezes Lloyd’s rule; a proof that the shift invariances used by Roth et al. are the only invariances of the trace form, whence the rest-length spring (not of negative type) has no dataset-independent PD representative; a proof that $1/n_c$ is the only size weight giving weighted kernel k -means objectives with fixed point weights ($3 \leq k \leq n - 2$); and exact rational certificates, on two explicit configurations, that a triangle-area three-body term, a cluster-mean rest length and a per-spring average are not pairwise objectives under either normalisation. These statements concern objective values, not minimisers; those on invariances and size weights are new here and not yet independently refereed. Computationally, on five datasets and six potentials the spring energies and the kernel objectives coincide to 1.1×10^{-11} on 6090 partitions, spring and kernel implementations of the same relaxation follow identical trajectories in 90/90 runs, and all 600 relaxation fixed points are kernel k -means fixed points. Dynamics on positions is left open. No claim is made about clustering performance.

1 Introduction

Physical metaphors recur in clustering: algorithms have been built on gravity [33], on spin models at finite temperature [3], on deterministic annealing of a free energy [21, 22], on quantum potentials [16], and on elastic nets [9, 34]. The research line to which this note belongs (LMI001,

“Mechanical analogies for data clustering”) studied formulations of a more elementary kind: points connected by springs, with tension and lifting terms, clustered by minimising an elastic energy. Its stated objective was to determine whether the mechanical component produces an effect distinguishable from matched kernel controls, and its screening found none. *The definitions of Section 2 are a reconstruction from the line’s public summary; no manuscript or code of the original screening was available to us, and the conclusions below apply to objectives of that pairwise form.* We likewise read “matched kernel control” as kernel k -means on the same affinity matrix with the same optimiser; this too is a reconstruction. The purpose of the note is to explain the null result rather than to repeat it, and the explanation is not new: for objectives of the pairwise form it is already contained in three literatures that the line’s summary does not mention. Pairwise data clustering [15] minimises exactly the per-particle spring energy for a dissimilarity matrix (by deterministic annealing, a finite-temperature treatment of the same energy); its reduction to k -means after a constant shift is the constant-shift embedding of Roth et al. [23]; the same objective is the ratio association that weighted kernel k -means optimises [7, 8]. For potentials of negative type, energy statistics supplies a dataset-independent kernel [29], and clustering by within-cluster energy is kernel k -means, optimised preferably by Hartigan’s method [10]. A control that matches the kernel therefore matches the per-particle objective exactly, and no experiment on objective values can separate them (experiments on optimisers can, Section 5); the summed energy is min-sum clustering, which is not a weighted kernel k -means objective (Proposition 4.3).

Contribution. The note collects these results for springs, with explicit constants, and draws the consequence for the research line (Section 3). Its own additions are modest: the explicit mechanism by which a diagonal shift freezes Lloyd’s rule while the energy and the single-move relaxation are unchanged (Proposition 3.10), quantified in E3; a proof that Lemma 3.1 lists all invariances of the trace form (Proposition 3.2, the converse of the shift invariance of 23), so that the rest-length spring, not of negative type (Proposition 3.5), has no dataset-independent PD representative (Corollary 3.6); the size weights that give weighted kernel k -means objectives (only $1/n_c$, Proposition 4.2; the total energy is not one even with data-dependent weights, Proposition 4.3); exact rational certificates that three specific functionals (a triangle-area three-body term, a cluster-mean rest length and a per-spring average) are not pairwise objectives under the per-particle or total normalisation (Section 4; these concern objective values, not minimisers); a reformulation of the dynamic open problem (Problem 4.4); and a computational confirmation of the identities with predefined criteria (Section 5). The results on invariances and size weights are checked by exact rank computations for small n , not yet by a referee. The new mathematical content is small; the note’s value, if any, is to make the null screening a corollary of known results.

Scope. The note is about objectives and their minimisers, not about clustering quality. We do not claim that any potential clusters real data well, and the adjusted Rand indices reported are context, not results. The datasets are standard synthetic ones and two small scikit-learn datasets [20]. Every number in the text is generated by the scripts described in Appendix A, with fixed seeds. Section 6 lists all claims with their status.

2 The pairwise spring family

Throughout, $X = (x_1, \dots, x_n)$ is a configuration in $\mathbb{R}^d$, $k \geq 2$ is fixed, and a *partition* $C = (C_1, \dots, C_k)$ is a partition of $[n]$ into k non-empty clusters, with sizes $n_c = |C_c|$ and indicator vectors $\mathbf{1}_c \in \{0, 1\}^n$. We write $D_{ij} = \|x_i - x_j\|$. “PSD” always refers to a matrix (positive semidefinite); a *kernel* on a set is called *positive definite* (PD) when all its Gram matrices are PSD, the convention of Berg et al. [2].

Definition 2.1 (Spring network, elastic energy, variants). A *spring network* is a triple (ϕ, L, w) of a *pair potential* $\phi: [0, \infty) \rightarrow \mathbb{R}$, a *lift* $L: \mathbb{R}^d \rightarrow \mathbb{R}^D$ (the identity by default), and a *normalisation* $w \in \{w_{\text{sp}}, w_{\text{tot}}\}$ with $w_{\text{sp}}(m) = 1/m$ (energy per particle of the cluster) and $w_{\text{tot}}(m) = 1$ (total energy). Its *within-cluster elastic energy* is

$$E_{\phi, L, w}(C) = \sum_{c=1}^k w(n_c) \sum_{\substack{i < j \\ i, j \in C_c}} \phi(\|Lx_i - Lx_j\|). \quad (1)$$

We write $A_{ij} = \phi(\|Lx_i - Lx_j\|)$ for the *spring matrix*, E_{sp} and E_{tot} for the two normalisations, and $\text{SSE}(C) = \sum_c \sum_{i \in C_c} \|x_i - \mu_c\|^2$, μ_c the centroid, for the k -means objective [18, 19]. The *released tension* of C is the energy of the springs that cross clusters, $T(C) = \sum_{c < c'} \sum_{i \in C_c, j \in C_{c'}} A_{ij}$; a tension formulation maximises T . An *anchored* network attaches every particle to a hub $y_c \in \mathbb{R}^D$ of its cluster, with energy $E^{\text{anch}}(C, y) = \sum_c \sum_{i \in C_c} \phi(\|Lx_i - y_c\|)$, minimised over the hubs as well as over C .

The potentials used below are: the Hooke spring with zero rest length, $\phi(d) = d^2$ (the spring constant is absorbed); the Hooke spring with rest length $r > 0$, $\phi(d) = (d - r)^2$; the string under constant tension, $\phi(d) = d$, whose per-particle energy is the within-cluster dispersion of energy statistics [10]; the saturating (breakable) spring, $\phi(d) = 1 - e^{-d^2/2\ell^2}$; the soft spring, $\phi(d) = \log(1 + d^2/\ell^2)$; and the Hooke spring on the paraboloid lift $L(x) = (x, \|x\|^2/s)$.

Definition 2.2 (Kernel k -means objective, trace form). For a symmetric matrix $K \in \mathbb{R}^{n \times n}$ and a partition C ,

$$J_K(C) = \text{tr } K - \sum_{c=1}^k \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c}. \quad (2)$$

When K is PSD with a Gram factorisation $K_{ij} = \langle \Phi_i, \Phi_j \rangle$, $J_K(C) = \sum_c \sum_{i \in C_c} \|\Phi_i - m_c\|^2$ with m_c the centroid of the Φ_i , $i \in C_c$: the k -means objective in feature space [6, 12, 28, 35]. For general symmetric K we take (2) as the definition.

Definition 2.3 (Negative type, Bernstein functions). A symmetric kernel ψ on a set S is *conditionally negative definite* (CND) if $\sum_{i,j} c_i c_j \psi(s_i, s_j) \leq 0$ for all finite families $s_i \in S$ and reals c_i with $\sum_i c_i = 0$. A function $\phi: [0, \infty) \rightarrow \mathbb{R}$ is *of negative type* if $(x, y) \mapsto \phi(\|x - y\|)$ is CND on $\mathbb{R}^d$ for every d . A function $\psi: [0, \infty) \rightarrow [0, \infty)$ is a *Bernstein function* if it is continuous, C^∞ on $(0, \infty)$, and $(-1)^{m-1} \psi^{(m)} \geq 0$ for all $m \geq 1$; equivalently (Lévy–Khintchine representation, 25, Thm. 3.2) $\psi(s) = a + bs + \int_{(0, \infty)} (1 - e^{-ts}) \mu(dt)$ with $a, b \geq 0$ and $\int \min(1, t) \mu(dt) < \infty$.

3 Results

3.1 Invariances of the trace form

Lemma 3.1 (What J does not see). *Let K be symmetric, C a partition into k non-empty clusters, $\sigma, a \in \mathbb{R}$ and $f \in \mathbb{R}^n$. Then*

$$J_{K+\sigma I}(C) = J_K(C) + \sigma(n - k), \quad J_{K+a\mathbf{1}\mathbf{1}^\top}(C) = J_K(C), \quad J_{K+f\mathbf{1}^\top + \mathbf{1}f^\top}(C) = J_K(C).$$

Proof. $\text{tr}(\sigma I) = \sigma n$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c / n_c = \sigma$ for each of the k clusters. $\text{tr}(a\mathbf{1}\mathbf{1}^\top) = an$ and $\sum_c an_c^2 / n_c = an$. $\text{tr}(f\mathbf{1}^\top + \mathbf{1}f^\top) = 2 \sum_i f_i$ and $\mathbf{1}_c^\top (f\mathbf{1}^\top + \mathbf{1}f^\top) \mathbf{1}_c / n_c = 2 \sum_{i \in C_c} f_i$, which sums to $2 \sum_i f_i$ over c . $\square$

In particular the minimisers of J_K over partitions into k clusters are unchanged by a diagonal shift, by a constant added to every entry, and by a separable term $f_i + f_j$. The lemma is elementary and not new: the diagonal shift is the constant-shift embedding of Roth et al. [23]

(a constant added to the off-diagonal dissimilarities) and the device of Dhillon et al. [7, 8] for making an indefinite affinity matrix PSD without changing the optimal partitions; the other two invariances connect the spring energy to Schoenberg’s theory of metric embeddings [26, 27] below.

Proposition 3.2 (Lemma 3.1 is complete). *Let $n \geq 3$, $2 \leq k \leq n - 1$ and $\Delta \in \mathbb{R}^{n \times n}$ symmetric. Then $J_\Delta(C)$ takes the same value on all partitions of $[n]$ into k non-empty clusters if and only if $\Delta = \sigma I + g\mathbf{1}\mathbf{1}^\top + \mathbf{1}g^\top$ for some $\sigma \in \mathbb{R}$, $g \in \mathbb{R}^n$, and then $J_\Delta \equiv \sigma(n - k)$. The pair (σ, g) is unique, so this space $V_{n,k}$ has dimension $n + 1$ and does not depend on k ; it is also the space of Δ with $J_\Delta(C) = \alpha + \beta(n - k)$ for all partitions with any number of clusters (and then $\alpha = 0$, $\beta = \sigma$). For $k \in \{1, n\}$ there is a single partition and every Δ qualifies. Equivalently, for $2 \leq k \leq n - 1$: $J_K - J_{K'}$ is constant on k -partitions iff the induced dissimilarities $D(K)_{ij} = K_{ii} + K_{jj} - 2K_{ij}$ and $D(K')$ differ by a constant off the diagonal, iff $P(K - K')P \in \mathbb{R}P$ with $P = I - \mathbf{1}\mathbf{1}^\top/n$. [proved here (elementary); proof in Appendix B]*

The proof reduces to zero diagonal and exchanges two points between clusters. Since $J_K(C) = \sum_c \frac{1}{n_c} \sum_{i < j \in C_c} D(K)_{ij}$, the pairwise-clustering cost of a dissimilarity matrix [15] determines the off-diagonal dissimilarities up to one additive constant: the converse of the shift invariance of Roth et al. [23]. We did not find it stated in the literature we know (not searched exhaustively). The dimension $n + 1$ is confirmed by exact ranks for $n \leq 7$ and all k (theory/check_invariances.py).

Corollary 3.3 (All representatives, and the minimal shift). *Let K be symmetric, $n \geq 3$, $2 \leq k \leq n - 1$. The symmetric K' with $J_{K'} - J_K$ constant on k -partitions are exactly $K' = K + \sigma I + g\mathbf{1}\mathbf{1}^\top + \mathbf{1}g^\top$. Such a K' can be chosen PSD iff $\sigma \geq \sigma_*(K) := \max\{-c^\top Kc : c \perp \mathbf{1}, \|c\| = 1\}$; for $\sigma \geq \sigma_*(K)$, $K' = P(K + \sigma I)P$ is one. [proved here (elementary)]; for $\sigma_* \geq 0$ the threshold is, in dissimilarity language, the minimal constant shift of Roth et al. [23] as we recall it (not checked against their text).*

Proof. The first sentence is Proposition 3.2 applied to $K' - K$. If $K' \succeq 0$ and $c \perp \mathbf{1}$, then $0 \leq c^\top K'c = c^\top Kc + \sigma\|c\|^2$, so $\sigma \geq \sigma_*(K)$. Conversely $P(K + \sigma I)P = K + \sigma I + g\mathbf{1}\mathbf{1}^\top + \mathbf{1}g^\top$ for some g (expand $P = I - \mathbf{1}\mathbf{1}^\top/n$ and absorb $\mathbf{1}\mathbf{1}^\top$ into g), and it is PSD iff $c^\top (K + \sigma I)c \geq 0$ for all $c \perp \mathbf{1}$. $\square$

3.2 Springs are kernels

Theorem 3.4 (Spring energies and kernel objectives; collected from the literature). *Let (ϕ, L, w) be a spring network with spring matrix A and let $K = -A$ (diagonal included, $K_{ii} = -\phi(0)$). For every partition C into k non-empty clusters:*

- (a) $E_{\phi, L, w}(C)$ depends on the data only through A .
- (b) (per-particle normalisation) $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n - k)\phi(0)$. For every $\sigma \geq -\lambda_{\min}(K)$ the matrix $K + \sigma I$ is PSD and $E_{\text{sp}}(C) = \frac{1}{2}J_{K + \sigma I}(C) + \frac{1}{2}(n - k)(\phi(0) - \sigma)$; hence $\arg \min_C E_{\text{sp}} = \arg \min_C J_{K + \sigma I}$, the k -means problem in the feature space of any Gram factorisation of $K + \sigma I$.
- (c) (Hooke) If $\phi(d) = d^2$ and L is the identity, $E_{\text{sp}}(C) = \text{SSE}(C)$ exactly.
- (d) (total normalisation and tension) $E_{\text{tot}}(C) + T(C) = \sum_{i < j} A_{ij}$ is independent of C ; minimising E_{tot} is maximising the released tension, which is the max- k -cut, equivalently min-sum k -clustering, problem with weights A_{ij} [24]. If all clusters have the same size, $E_{\text{tot}} = (n/k)E_{\text{sp}}$.
- (e) (converse) Every kernel k -means objective is a Hooke energy: if K is PSD and $K = \Phi\Phi^\top$, then $J_K(C) = E_{d^2, \Phi, \text{sp}}(C)$, the per-particle Hooke energy of the lifted configuration $(\Phi_1, \dots, \Phi_n)$.

(f) (universal kernel) If ϕ is of negative type, then

$$\widetilde{K}(x, y) = \phi(\|x\|) + \phi(\|y\|) - \phi(\|x - y\|) - \phi(0)$$

is a PD kernel on $\mathbb{R}^D$ that does not depend on the dataset (given the scales ℓ, r, s that enter ϕ and L), and $E_{\text{sp}}(C) = \frac{1}{2}J_{\widetilde{K}}(C) + \frac{1}{2}(n - k)\phi(0)$ where $\widetilde{K}$ is evaluated on the lifted points. If Φ is a feature map of $\widetilde{K}$ and $\tilde{\Phi} = \Phi/\sqrt{2}$, then $\|\tilde{\Phi}(x) - \tilde{\Phi}(y)\|^2 = \phi(\|x - y\|) - \phi(0)$ and

$$E_{\text{sp}}(C) = E_{d^2, \tilde{\Phi} \circ L, \text{sp}}(C) + \frac{1}{2}(n - k)\phi(0) :$$

up to the constant, the spring energy is the per-particle Hooke energy of the lifted configuration $(\tilde{\Phi}(Lx_i))_i$ (the lift $\tilde{\Phi} \circ L$ takes values in a possibly infinite-dimensional feature space; restricted to the n lifted points it has finite rank, so Definition 2.1 applies with finite D). Without the rescaling, E_{sp} is half the Hooke energy of $(\Phi(Lx_i))_i$ plus the constant.

Status. The theorem collects known results; we state it for springs because the explicit constants are what E1 tests. (a), (d) elementary (unlike E_{sp} , E_{tot} is not a weighted kernel k -means objective for non-negative springs, Proposition 4.3); (c) classical; (e) the feature-space reading of k -means [12, 35] [classical]. (b) [literature]: with zero diagonal, E_{sp} is the pairwise-clustering cost of Hofmann and Buhmann [15] for the dissimilarities A_{ij} , and its reduction to k -means after a constant shift is the constant-shift embedding of Roth et al. [23]; equivalently, $\sum_c \mathbf{1}_c^\top A \mathbf{1}_c / n_c$ is the ratio association of A , whose maximisation is kernel k -means with $K = \sigma I + A$ [7, 8], and minimising E_{sp} maximises the ratio association of $-A$, i.e. uses $K = \sigma I - A$. Specific to springs is only the constant $\frac{1}{2}(n - k)\phi(0)$. (f) [literature]: $\frac{1}{2}\widetilde{K}$ is the distance-induced kernel of Sejdinovic et al. [29] for the semimetric of negative type $\rho(x, y) = \phi(\|x - y\|) - \phi(0)$ with base point 0 (it is ≥ 0 because the CND test on two points forces $\phi(d) \geq \phi(0)$), $\tilde{\Phi}$ is its feature map, $E_{\text{sp}} - \frac{1}{2}(n - k)\phi(0)$ is the within-cluster energy dispersion of energy statistics for ρ , and its equivalence with kernel k -means is in França et al. [10]; we include the short argument.

Proof. (a) is the definition. (b) Since $\sum_{i < j \in C_c} A_{ij} = \frac{1}{2}(\mathbf{1}_c^\top A \mathbf{1}_c - n_c \phi(0))$,

$$E_{\text{sp}}(C) = \frac{1}{2} \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} - \frac{1}{2}k\phi(0), \quad \text{and} \quad \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} = - \sum_c \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c} = J_K(C) - \text{tr } K,$$

with $\text{tr } K = -n\phi(0)$, so $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n - k)\phi(0)$. The shifted form follows from Lemma 3.1, and $K + \sigma I \succeq 0$ for $\sigma \geq -\lambda_{\min}(K)$. (c) For any finite set P with centroid μ , $\sum_{i, j \in P} \|x_i - x_j\|^2 = 2|P| \sum_{i \in P} \|x_i - \mu\|^2$ (expand $\|(x_i - \mu) - (x_j - \mu)\|^2$; the cross terms vanish). Halving and dividing by $|P|$ gives $\frac{1}{n_c} \sum_{i < j \in C_c} d_{ij}^2 = \sum_{i \in C_c} \|x_i - \mu_c\|^2$; sum over c . (d) Every pair $\{i, j\}$ is either within a cluster or across, so $E_{\text{tot}} + T$ is the total spring energy; the equal-size statement is $w_{\text{tot}} = (n/k)w_{\text{sp}}$. (e) By (b) with $\phi(0) = 0$ applied to the configuration Φ : $E_{d^2, \Phi, \text{sp}}(C) = \frac{1}{2}J_{-D_{\Phi}^{(2)}}(C)$, where $(D_{\Phi}^{(2)})_{ij} = K_{ii} + K_{jj} - 2K_{ij}$, so $-D_{\Phi}^{(2)} = 2K - f\mathbf{1}^\top - \mathbf{1}f^\top$ with $f_i = K_{ii}$, and Lemma 3.1 gives $\frac{1}{2}J_{2K}(C) = J_K(C)$. (f) Positive definiteness of $\widetilde{K}$ is the lemma relating CND and PD kernels [2, Ch. 3, §2, Lemma 2.1]: for points $x_1, \dots, x_m$ and reals c_i , apply the CND inequality to the family augmented by $x_0 = 0$, $c_0 = -\sum_{i \geq 1} c_i$; substituting c_0 it reads $\sum_{i, j \geq 1} c_i c_j [\phi(\|x_i - x_j\|) - \phi(\|x_i\|) - \phi(\|x_j\|) + \phi(0)] \leq 0$, i.e. $\sum_{i, j} c_i c_j \widetilde{K}(x_i, x_j) \geq 0$. On the dataset, $\widetilde{K} = K + f\mathbf{1}^\top + \mathbf{1}f^\top - \phi(0)\mathbf{1}\mathbf{1}^\top$ with $f_i = \phi(\|Lx_i\|)$, so $J_{\widetilde{K}} = J_K$ by Lemma 3.1 and the identity follows from (b). Next, $\|\Phi(x) - \Phi(y)\|^2 = \widetilde{K}(x, x) + \widetilde{K}(y, y) - 2\widetilde{K}(x, y) = 2\phi(\|x - y\|) - 2\phi(0)$, so $\|\tilde{\Phi}(x) - \tilde{\Phi}(y)\|^2 = \phi(\|x - y\|) - \phi(0)$. Finally (e) applied to $\widetilde{K}$ gives $J_{\widetilde{K}}(C) = E_{d^2, \Phi \circ L, \text{sp}}(C)$, and scaling a configuration by $1/\sqrt{2}$ halves every squared distance, so $\frac{1}{2}J_{\widetilde{K}}(C) = E_{d^2, \tilde{\Phi} \circ L, \text{sp}}(C)$. For Hooke springs $\widetilde{K}(x, y) = 2\langle x, y \rangle$, $\Phi(x) = \sqrt{2}x$ and $\tilde{\Phi}(x) = x$. $\square$

Proposition 3.5 (Which springs are of negative type). (i) $\phi(d) = d^2$ is of negative type: for $\sum_i c_i = 0$, $\sum_{i, j} c_i c_j \|x_i - x_j\|^2 = -2\|\sum_i c_i x_i\|^2 \leq 0$. [classical]

(ii) If N is a CND kernel with $N \geq 0$ and ψ is a Bernstein function, then $\psi \circ N$ is CND. [literature]

(iii) Consequently $\phi(d) = \psi(d^2)$ is of negative type for every Bernstein ψ ; this covers $d^{2\alpha}$ for $0 < \alpha \leq 1$ (in particular d), $\log(1 + d^2/\ell^2)$, $1 - e^{-d^2/2\ell^2}$ and $d^2/(1 + d^2/\ell^2)$. The rest-length potential $(d - r)^2$ is not of negative type. [proved here (elementary)]

Proof sketch. (i) Expand $\|x_i - x_j\|^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle$; the first two terms vanish against $\sum c_i = 0$. (ii) is proved in Berg et al. [2, Ch. 3, §2] (Schoenberg’s theorem is Theorem 2.2 there; the composition with Bernstein functions follows from it and the representation in Definition 2.3), see also Schilling et al. [25]; we do not vouch for the exact numbering of the composition result. The argument is: by Schoenberg’s theorem [27] N is CND iff e^{-tN} is PD for all $t > 0$, so $1 - e^{-tN}$ is CND (a constant kernel is CND-neutral and the negative of a PD kernel is CND); write $\psi(N) = a + bN + \int (1 - e^{-tN})\mu(dt)$ and note that non-negative combinations and pointwise limits of CND kernels are CND. (iii) The listed ψ are Bernstein: s^α ($0 < \alpha \leq 1$), $\log(1 + s/\ell^2)$, $1 - e^{-s/2\ell^2}$, $s/(1 + s/\ell^2)$ all have completely monotone derivatives. If ϕ is of negative type, the CND inequality for two points with $c = (1, -1)$ gives $2\phi(0) - 2\phi(d) \leq 0$, i.e. $\phi(d) \geq \phi(0)$ for all d ; for $(d - r)^2$, $\phi(r) = 0 < r^2 = \phi(0)$. *Rest length.* Theorem 3.4(b) still applies to $(d - r)^2$: its spring energy is the kernel k -means objective of $-(D - r)^{(2)}$ after a diagonal shift, and its minimisers are feature-space k -means minimisers for that dataset. What fails is a dataset-independent PD kernel. $\widetilde{K}$ is not PD, since $\widetilde{K}(x, x) = 2(\phi(\|x\|) - \phi(0)) = -2r^2 < 0$ when $\|x\| = r$. Nor is any kernel $\widetilde{K}(x, y) + f(x) + f(y) + a + \sigma[x = y]$ with f, a, σ fixed: for the collinear points $0, su, 2su, 3su$ (u a unit vector, $s > 0$) and $c = (1, -1, -1, 1)$, the terms in f and a vanish against $\sum_i c_i = 0$ and the quadratic form of the Gram matrix is $\sum_{i,j} c_i c_j (-\phi(d_{ij})) + 4\sigma = -8rs + 4\sigma < 0$ once $s > \sigma/2r$. By Proposition 3.2 this class contains every dataset-independent representative, so none is PD (Corollary 3.6). Accordingly, E1 finds $\widetilde{K}$ indefinite for this potential on every dataset (Section 5). $\square$

Corollary 3.6 (The rest-length spring has no dataset-independent PD representative). *Let $\phi: [0, \infty) \rightarrow \mathbb{R}$ and $A[X]_{ij} = \phi(\|x_i - x_j\|)$. Call a symmetric $\kappa: \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$ a dataset-independent representative of ϕ if for every set $X \subset \mathbb{R}^D$ of $n \geq 3$ distinct points and some $k \in \{2, \dots, n - 1\}$ (equivalently, by Proposition 3.2, every such k), $J_{\kappa[X]} - J_{-A[X]}$ is constant on the partitions of X into k clusters.*

(a) κ is a dataset-independent representative iff there are $g: \mathbb{R}^D \rightarrow \mathbb{R}$ and $\sigma \in \mathbb{R}$ with $\kappa(x, y) = -\phi(\|x - y\|) + g(x) + g(y)$ for $x \neq y$ and $\kappa(x, x) = -\phi(0) + \sigma + 2g(x)$. This is the class $\widetilde{K} + f(x) + f(y) + a + \sigma[x = y]$ of Proposition 3.5.

(b) For $\phi(d) = (d - r)^2$, $r > 0$, no dataset-independent representative is PD. Quantitatively, on $X_s = \{0, su, 2su, 3su\}$ ($\|u\| = 1$) the minimal shift of Corollary 3.3 satisfies $\sigma_*(-A[X_s]) \geq 2rs$, which is unbounded in s .

(c) (b) still holds if $J_{\kappa[X]}$ is only required to equal $\lambda_X J_{-A[X]}$ plus a constant, with a factor $\lambda_X > 0$ that may depend on X .

[proved here (elementary); proof of (c) in Appendix B]

Proof of (a) and (b). (a) If: Lemma 3.1. Only if: by Proposition 3.2, for each X there is a unique (σ_X, g_X) with $\kappa[X] + A[X] = \sigma_X I + g_X \mathbf{1}^\top + \mathbf{1} g_X^\top$. If $X \subseteq Y$, the principal submatrix of Y ’s decomposition is a decomposition for X , so by uniqueness $\sigma_X = \sigma_Y$ and $g_X = g_Y|_X$. Comparing any X, X' through $X \cup X'$ shows that $\sigma_X = \sigma$ is one constant and that $g(x) := g_X(x)$ does not depend on the set $X \ni x$. (b) For $c = (1, -1, -1, 1)$ on X_s the terms in g vanish against $\sum_i c_i = 0$, and $\sum_{i,j} c_i c_j \kappa(x_i, x_j) = -\sum_{i,j} c_i c_j \phi(d_{ij}) + 4\sigma = -8rs + 4\sigma < 0$ for $s > \sigma/2r$ (the computation in the proof of Proposition 3.5), so κ is not PD. Also $c/2$ is a unit vector orthogonal to $\mathbf{1}$ with $(c/2)^\top A[X_s](c/2) = 2rs$, whence $\sigma_* \geq 2rs$. $\square$

Remark 3.7 (Same minimisers instead of same values). (a) If a fixed symmetric Δ satisfies $\arg \min J_{K+\Delta} = \arg \min J_K$ for every PD (a fortiori every symmetric) K , then $\Delta \in V_{n,k}$: otherwise $K = \lambda I - \Delta$ with λ large is PD, $J_{K+\Delta} = J_{\lambda I}$ is constant, and $\arg \min J_K = \arg \max J_{\Delta}$ is a proper subset. As uniform transformations of the matrix, “same minimisers” adds nothing to Proposition 3.2 [proved here (one line)]. (b) On a fixed dataset the weaker notion is vacuous: $K + \sigma I$ ($\sigma \geq \sigma_*$) is PSD with the same minimisers, and so is the block matrix $\sum_c \mathbf{1}_c \mathbf{1}_c^\top$ of a unique minimiser. (c) Open: whether some dataset-independent PD kernel κ has $\arg \min J_{\kappa[X]} = \arg \min J_{-A[X]}$ for every dataset and every k , for the rest-length spring. Corollary 3.6(c) excludes positive affine reparametrisations; three-point sets with $k = 2$, where $J_K(\{i\}, \{j, l\}) = \frac{1}{2} D(K)_{jl}$, show that the pair closest in the feature metric of κ must be the pair whose distance is closest to r . The linear algebra used here does not reach ordinal conditions of this kind.

Remark 3.8 (Lifting and anchoring). (a) $E_{\phi, L, w}(X) = E_{\phi, \text{id}, w}(LX)$, so a lift replaces A by $\phi(\|Lx_i - Lx_j\|)$ and nothing else; for Hooke springs a lift is exactly the choice of the kernel $\langle Lx, Ly \rangle$, and by Theorem 3.4(e) every PD kernel arises this way on finite data [trivial]. (b) For Hooke springs, $\min_y E^{\text{anch}}(C, y) = E_{\text{sp}}(C) = \text{SSE}(C)$, attained at the centroids (Theorem 3.4(c) and the fact that the centroid minimises the sum of squared distances), and the alternating equilibration “hubs move to the equilibrium of their springs; particles re-attach to the hub of least energy” is Lloyd’s algorithm [18] by definition. With $\phi(d) = d$ the anchored objective is the k -median objective (geometric medians as hubs) [classical]; we make no claim about whether such anchored objectives are of the form (1).

3.3 Relaxation of the spring system

The mechanical way to minimise (1) is to let the system relax: one particle at a time jumps to the cluster in which its springs store less energy, or, at finite temperature, jumps with Metropolis probability. On assignments this is a known algorithm.

Proposition 3.9 (Zero-temperature relaxation is Hartigan’s method). *Let K' be any PSD matrix with $J_{K'} = J_K + \text{const}$ on partitions (for instance $K + \sigma I$ or, for ϕ of negative type, $\bar{K}$; by Corollary 3.3 these are exactly the PSD matrices $K + \sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top$), and write $\delta_{K'}(i, c) = K'_{ii} - \frac{2}{n_c}(K'\mathbf{1}_c)_i + \frac{1}{n_c^2}\mathbf{1}_c^\top K'\mathbf{1}_c$ for the squared feature distance from Φ_i to the centroid of cluster c . Moving particle i from its cluster a ($n_a \geq 2$) to cluster c changes E_{sp} by*

$$\Delta E = \frac{1}{2} \left[\frac{n_c}{n_c+1} \delta_{K'}(i, c) - \frac{n_a}{n_a-1} \delta_{K'}(i, a) \right], \quad (3)$$

and the right-hand side does not depend on the representative K' . Consequently:

- (i) the greedy single-particle relaxation of the spring energy is Hartigan’s method [13, 14] applied to the kernel k -means objective;
- (ii) every partition that is stable under single-particle moves is Voronoi-stable in the feature space of every PSD representative K' , i.e. each Φ_i is assigned to a nearest centroid, so it is a fixed point of Lloyd’s algorithm for K' with ties broken in favour of the current cluster (on the relation between such fixed points and local optimality see 30);
- (iii) the converse fails: $X = \{0, 2, 3.4, 3.6\} \subset \mathbb{R}$, $k = 2$, $C = \{\{0, 2\}, \{3.4, 3.6\}\}$ is Voronoi-stable with $\text{SSE} = 2.02$, while moving the point 2 gives $\text{SSE} = 1.52$.

Proof. Write $S_{ic} = (K'\mathbf{1}_c)_i$ and $Q_c = \mathbf{1}_c^\top K'\mathbf{1}_c$. Removing i from a changes Q_a/n_a to $(Q_a - 2S_{ia} + K'_{ii})/(n_a - 1)$, adding it to c changes Q_c/n_c to $(Q_c + 2S_{ic} + K'_{ii})/(n_c + 1)$, and $\Delta J = -\Delta \sum_c Q_c/n_c$. Expanding both differences and comparing with $\delta_{K'}$ gives $\Delta J = \frac{n_c}{n_c+1} \delta_{K'}(i, c) - \frac{n_a}{n_a-1} \delta_{K'}(i, a)$, an algebraic identity valid for any symmetric K' ; $\Delta E = \frac{1}{2} \Delta J$ by Theorem 3.4(b). Independence

of the representative: if $J_{K'} = J_K + \text{const}$ on partitions then $\Delta J_{K'} = \Delta J_K$ for every move, and the identity holds for both matrices, so the right-hand side of (3) is the same for every such K' . Explicitly, under $K' \mapsto K' + \sigma I$, $\delta(i, c)$ increases by $\sigma(1 + 1/n_c)$ for $i \notin C_c$ and by $\sigma(1 - 1/n_a)$ for $i \in C_a$, and both terms of (3) increase by σ ; under the other two invariances of Lemma 3.1, δ itself is unchanged. (i) is the definition of Hartigan’s move. (ii) If no move decreases E then $\frac{n_c}{n_c+1}\delta(i, c) \geq \frac{n_a}{n_a-1}\delta(i, a)$ for all $c \neq a$, and since $\frac{n_c}{n_c+1} < 1 < \frac{n_a}{n_a-1}$ this forces $\delta(i, c) \geq \delta(i, a)$ (not necessarily strictly, whence the tie rule). (iii) Centroids 1 and 3.5; the point 2 is at distance 1 from the first and 1.5 from the second; after the move the clusters $\{0\}$ and $\{2, 3.4, 3.6\}$ (centroid 3) have $\text{SSE} = 0 + 1 + 0.16 + 0.36 = 1.52$; by (3) with $K' = \tilde{K} = 2\langle x, y \rangle$, $\Delta E = \frac{1}{2}(\frac{2}{3} \cdot 2 \cdot 1.5^2 - 2 \cdot 2 \cdot 1^2) = -0.5$ (equivalently, the bracket of (3), which is $\Delta J_{K'}$ for any symmetric K' , with $K' = \langle x, y \rangle$ and $J_{\langle x, y \rangle} = \text{SSE}$). $\square$

Status. (i) [literature]: Hartigan’s method in kernel space is the kernel k -groups algorithm of França et al. [10], there derived for energy statistics; the representative-free form (3) is elementary. (ii) is the kernel form of the Euclidean inclusion of Telgarsky and Vattani [31] [proved here]; we could not check whether França et al. [10] state it and claim no priority. (iii) is the standard type of example. The consequence for the research line is that “spring relaxation” cannot converge to anything outside the set of kernel k -means fixed points.

Proposition 3.10 (Lloyd’s rule is representation-dependent). *For $K' = K + \sigma I$, $\delta_{K'}(i, c) = \delta_K(i, c) + \sigma(1 + \frac{1-2\cdot\mathbb{I}_{i \in C_c}}{n_c})$: the shift adds $\sigma(1 + 1/n_c)$ to the squared distance from i to every other cluster c and only $\sigma(1 - 1/n_a)$ to the distance to its own cluster a . Hence the Voronoi rule “assign i to $\arg \min_c \delta_{K'}(i, c)$ ” favours staying, and as $\sigma \rightarrow \infty$ every partition becomes Voronoi-stable, while E_{sp} and the single-move dynamics are unchanged. [proved here (two lines); explicit form of an effect noted by Dhillon et al. [7, 8]]*

Proof. $(\sigma I \mathbf{1}_c)_i = \sigma[i \in C_c]$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c = \sigma n_c$; substitute in δ . $\square$

Dhillon et al. [7, 8] already discuss how the diagonal shift affects the practical performance of kernel k -means; the proposition is the explicit mechanism and E3 quantifies it. So “kernel k -means with Lloyd’s iteration” is not one algorithm for a spring energy but one per PSD representative; the single-move relaxation is the representative-free one. For springs of negative type the natural representative is $\tilde{K}$ of Theorem 3.4(f), which for Hooke springs is $2\langle x, y \rangle$ and reproduces ordinary Lloyd k -means; the experiments use it, and use the naively shifted $-A + \sigma I$ only as a control (E3).

4 What leaves the family

Theorem 3.4 covers every objective that is a sum over within-cluster *pairs* of a fixed function of the data, weighted by a function of the cluster size in $\{1, 1/n_c\}$. An ingredient can escape only by breaking one of these three features: the sum over pairs (many-body terms), the fixedness of the pair function (parameters that depend on the assignment), or the normalisation. A fourth route replaces the minimisation over partitions by a dynamics on positions. For the first three, specific instances can be certified exactly.

Theorem 4.1 (Exact separation on explicit configurations; computer-assisted). *Let P_1 and P_2 be the configurations of 7 points in $\mathbb{Q}^2$*

$$\begin{aligned} P_1 &= \{(5, 10), (6, 2), (4, 4), (2, 2), (3, 7), (0, 6), (8, 8)\}, \\ P_2 &= \{(0, 2), (7, 4), (8, 7), (5, 0), (2, 8), (2, 3), (4, 10)\}, \end{aligned}$$

drawn once from a dedicated generator (seed 20260932), and let $k = 2$, so that each has 63 partitions. Call a partition function F w -pairwise if $F(C) = \sum_c w(n_c) \sum_{i < j \in C_c} A_{ij} + a$ for some

symmetric A and constant a (22 unknowns). Then, with squared distances $s_{ij} = \|x_i - x_j\|^2$, the following holds on both configurations (in parentheses the relative residual $\|b - \Pi b\|/\|b\|$ on P_1 ; P_2 , where $b \in \mathbb{Q}^{63}$ lists the values of F and Π is the orthogonal projection onto the w -pairwise functions):

partition function	w_{sp} -pairwise?	w_{tot} -pairwise?
$\sum_c \frac{1}{n_c} \sum d_{ij}^2$ (Hooke, per particle)	yes (0; 0)	no (0.017; 0.015)
$\sum_c \sum d_{ij}^2$ (Hooke, total)	no (0.179; 0.179)	yes (0; 0)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho)^2$, fixed ρ	yes (0; 0)	no (0.026; 0.030)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho_c)^2$, $\rho_c = \text{cluster mean}$	no (0.143; 0.157)	no (0.109; 0.122)
$\sum_c \frac{1}{n_c} \sum_{i < j < l} 4 \text{area}(ijl)^2$ (three-body)	no (0.331; 0.328)	no (0.039; 0.037)
$\sum_c \binom{n_c}{2}^{-1} \sum d_{ij}^2$ (per spring)	no (0.185; 0.180)	no (0.125; 0.121)

where ρ is the mean of all s_{ij} , ρ_c the mean of s_{ij} over the pairs of cluster c , and $\text{area}(ijl)$ the area of the triangle $x_i x_j x_l$. Each row is one function F , tested against both families. $\square$

Proof. For fixed w the w -pairwise functions form a linear subspace of $\mathbb{R}^{63}$ spanned by the columns of an explicit rational design matrix (rank 21 for w_{sp} , 22 for w_{tot} , on both configurations). Membership is decided by exact least squares in rational arithmetic (`fractions.Fraction`): a zero residual exhibits A and a ; a non-zero residual is a certificate that F is not in the span. The script `experiments/identity_check.py` (E4) performs the computation; the relative residuals are reported in the table. $\square$

The two positive controls (Hooke and fixed rest length under their own normalisation, total Hooke under w_{tot}) confirm that the test is not blind, and the yes/no pattern agrees on the two configurations in 12 of the 12 cells (a sanity check: one configuration suffices logically). The table says that the two normalisations are different objectives (each fails to be pairwise under the other’s weight), and that the three functionals tested (rows 4–6: the cluster-mean rest length and the three-body term, both with the per-particle weight $1/n_c$ inside F , and the per-spring average) are neither w_{sp} - nor w_{tot} -pairwise; total-weight versions of rows 4–5 were not tested. The per-spring average *is* a pairwise sum with the size weight $\binom{n_c}{2}^{-1}$; the certificate only excludes the weights 1 and $1/n_c$. By Cauchy–Binet applied to the rows $(1, x_i^\top)$, $\sum_{i < j < l \in C} 4 \text{area}(ijl)^2 = |C| \det S_C$ with S_C the scatter matrix of C , so the three-body row is $\sum_c \det S_c$, a quartic in the data. The theorem certifies these three functionals, not the classes “many-body terms” or “assignment-dependent rest lengths”, and the class statements would be false: restricted to partitions whose clusters all have at least three points, $\sum_c w(n_c)/(n_c - 2) \sum_{i < j < l \in C_c} (s_{ij} + s_{jl} + s_{il}) = \sum_c w(n_c) \sum_{i < j \in C_c} s_{ij}$.

Values, not minimisers. The certificate concerns objective values up to an additive constant. It does not show that the minimisers differ from those of every pairwise objective: with A free, every partition is the unique minimiser of some w_{tot} -pairwise objective ($A_{ij} = -1$ within its blocks, 0 otherwise), and a strictly increasing non-affine transform of a pairwise objective in general fails the test while keeping its minimisers. Whether these ingredients can produce an *observable* mechanical effect is therefore a separate question that the theorem does not answer.

Size weights. The normalisation is the third route above. For point weights $\omega \in (0, \infty)^n$ fixed before the partition is chosen and a symmetric K , the *weighted kernel k-means* objective [6, 8] is $J_K^\omega(C) = \sum_i \omega_i K_{ii} - \sum_c \omega(C_c)^{-1} \sum_{i,j \in C_c} \omega_i \omega_j K_{ij}$, $\omega(S) = \sum_{i \in S} \omega_i$, which for $K = \Phi \Phi^\top$ is $\sum_c \sum_{i \in C_c} \omega_i \|\Phi_i - m_c\|^2$ with m_c the ω -weighted centroid [literature]. For a size weight w write $F_A^w(C) = \sum_c w(n_c) \sum_{i < j \in C_c} A_{ij}$; only $w(m)$, $2 \leq m \leq n - k + 1$, enters. Two facts are used. First, $\mathcal{G}_\omega = \{J_K^\omega : K \text{ symmetric}\}$ is a linear space of functions of the partition, of dimension at most $\binom{n}{2}$ (the n -dimensional space $\{g\mathbf{1}^\top + \mathbf{1}g^\top\}$ is in the kernel of $K \mapsto J_K^\omega$), and it contains the constants ($J_K^\omega \equiv n - k$ for $K = \text{diag}(1/\omega_i)$); so “up to a constant and a sign or positive factor” adds nothing. Second, F_A^w is linear in A and depends only on its off-diagonal part.

Proposition 4.2 (Size weights inside weighted kernel k -means, fixed point weights). *Let $n \geq 5$, $3 \leq k \leq n - 2$ and $w(2) \neq 0$. The following are equivalent:*

- (i) *there is $\omega \in (0, \infty)^n$ such that for every symmetric A there is a symmetric K with $F_A^\omega - J_K^\omega$ constant on k -partitions;*
- (ii) *the same for every Hooke spring matrix $A_{ij} = (x_i - x_j)^2$ of n distinct points $x_i \in \mathbb{R}$;*
- (iii) *$w(m) = 2w(2)/m$ for $m = 2, \dots, n - k + 1$.*

Under (iii) one can take $\omega = \mathbf{1}$ and $K = -w(2)A$ (zero diagonal). For $k = 2$ and $n \geq 4$, (i) implies that $h(a) + h(n - a)$ does not depend on $a \in \{1, \dots, n - 1\}$, where $h(m) = w(m)\binom{m}{2}$, provided $w(n - 1) \neq 0$ and $w(2) + w(n - 2) \neq 0$; for $n = 4$ this is again (iii). [proved here; proof in Appendix B]

The proof counts dimensions: under (i) the $\binom{n}{2}$ pair functions and the constant lie in $\mathcal{G}_\omega$, so a non-zero combination is constant, and the exchange argument of Proposition 3.2, weighted by w , forces $w(m)\binom{m}{2} = (m - 1)w(2)$. Taking n large, a size weight that works for all n and k is $w(m) \propto 1/m$ for all $m \geq 2$. The case $k = 2$ with $h(a) + h(n - a)$ constant and $w \not\propto 1/m$ is not settled by the dimension count. Point weights that depend on A , as in the normalised-cut equivalence of Dhillon et al. [6, 8], are a weaker requirement; for the total energy they do not help:

Proposition 4.3 (The total energy is not weighted kernel k -means). *Let $2 \leq k \leq n - 2$ and let A be symmetric with $A_{ij} \geq 0$ for $i \neq j$, not all zero (for instance the spring matrix of distinct points for any ϕ with $\phi(d) > 0$ for $d > 0$). Then for no $\omega \in (0, \infty)^n$, possibly depending on A , and no symmetric K is $E_{\text{tot}} - J_K^\omega$ constant on k -partitions. If $3 \leq k \leq n - 2$ or $(n, k) = (4, 2)$, the same holds for F_A^ω whenever $w(3) > 0$ and $w(3) \geq w(2)$. [proved here; proof in Appendix B]*

The proof exhibits, for each ω , a signed measure ν on seven partitions (the two-block splits of four groups of points) that annihilates $\mathcal{G}_\omega$ and the constants but pairs positively with E_{tot} . It excludes exact representation for every ω but gives no uniform bound on the distance from E_{tot} to $\mathcal{G}_\omega$, which in small numerical examples becomes small as ω degenerates. For $w = 1/m$ the pairing vanishes at $\omega = \mathbf{1}$, as it must. For decreasing weights with $w(3) < w(2)$ (e.g. $1/m^2$, $\binom{m}{2}^{-1}$) it changes sign on $(0, \infty)^4$, so the certificate does not apply, and we do not know whether data-dependent weights can represent them (Proposition 4.2 still excludes fixed weights). As functions of the partition (values up to a constant, not minimisers: with K free every partition is the unique minimiser of some kernel objective, Remark 3.7(b)), max- k -cut and min-sum k -clustering (Theorem 3.4(d)) are thus not weighted kernel k -means objectives for non-negative spring matrices, unlike ratio association and normalised cut [8]. We know of no statement of Propositions 4.2–4.3 in the literature (not searched exhaustively).

The dynamic route. Several physics-inspired algorithms do not minimise an energy over partitions but move the *positions*: gravitational clustering [33], mean shift [5, 11] (query points ascend a fixed kernel density estimate) and its blurring variant, in which all data points move [4], quantum clustering [16], and the finite-temperature correlations of a Potts model [3]. Their output is a function of the trajectory, and Theorem 3.4 says nothing about it.

Open problem 4.4 (Dynamic mechanical clustering). Let $V: (0, \infty) \rightarrow \mathbb{R}$ be a pair potential, let $z(t)$ solve $\dot{z}_i = -\sum_{j \neq i} V'(\|z_i - z_j\|) \frac{z_i - z_j}{\|z_i - z_j\|}$ with $z(0) = X$, and let $P_{T,\varepsilon}(X)$ be the single-linkage partition of $z(T)$ at threshold ε . Characterise the triples (V, T, ε) for which there exist a fixed function $a: [0, \infty) \rightarrow \mathbb{R}$ and a normalisation w such that, for every X , $P_{T,\varepsilon}(X)$ is the *unique* minimiser of $\sum_c w(n_c) \sum_{i < j \in C_c} a(\|x_i - x_j\|)$ over partitions into the number of clusters it produces.

Remark 4.5 (Why “unique”, and how a counterexample could be certified). Without uniqueness the problem is trivial: for $a \equiv 0$ every partition is a minimiser (for $w = 1/n_c$, also for every constant a). With uniqueness, a single configuration whose pairwise distances are all distinct is never a counterexample: $a = -1$ on the distances inside the blocks of $P_{T,\varepsilon}(X)$ and $a = 0$ on the others makes it the unique minimiser for $w \in \{1, 1/n_c\}$. A counterexample needs repeated distances, within one configuration or across configurations that share distances (e.g. subsets of a lattice). For finitely many configurations it is a finite system of strict linear inequalities in the values of a on the common distance set (scaled to $F(C) - F(P) \geq 1$ for $C \neq P$), whose infeasibility a Farkas certificate over $\mathbb{Q}$ certifies exactly. We have not carried this out. [proved here; not yet checked by an independent referee]

For the Gaussian attraction $V(d) = -e^{-d^2/2h^2}$ the flow is a continuous-time, unnormalised form of Gaussian *blurring* mean shift [4], not of standard mean shift. We expect no such pair (a, w) to exist for generic (T, ε) , but have not tested it. Whatever the answer, a mechanical proposal of this kind has to be compared with (blurring) mean shift, not with k -means, and the comparison must involve the trajectory, not the energy minimum.

5 Computational confirmation

Data and potentials. Five datasets: blobs, two moons and two circles from scikit-learn’s generators (300 points each in $\mathbb{R}^2$; `make_blobs` with three centres and std 0.9, $k = 3$; `make_moons` with noise 0.06 and `make_circles` with factor 0.45 and noise 0.05, $k = 2$), a random subset of 300 digits from `load_digits` (64 features scaled to $[0, 1]$, $k = 10$) and the standardised iris data (150 points, 4 features, $k = 3$). The six potentials of Section 2, with scales set from the data before any run: ℓ is the median distance to the seventh nearest neighbour, r the tenth percentile of all pairwise distances, s the median of $\|x\|^2$. In all, $5 \times 6 = 30$ configurations. Seeds: 20260930 (E1, E2), 20260931 (E3) and 20260932 (E4).

E1: identical objective values. *Criterion (predefined):* for every tested partition and every potential, $|E_{\text{sp}} - (\frac{1}{2}J_{K'} + \text{const})| \leq 10^{-9} \max(1, |E_{\text{sp}}|)$ with the constant of Theorem 3.4, for the three representatives $K' \in \{-A, -A + \sigma I, \widetilde{K}\}$; pass means 100%. The partitions are 200 uniformly random ones per configuration, the outputs of Lloyd and of the single-move relaxation on $\widetilde{K}$ (minimally shifted when indefinite), and the ground-truth labels. *Result:* 19285/19285 checks passed, worst relative error 1.1×10^{-11} (the shifted representative uses σ up to 4.4×10^4 , so its identity involves cancellation); the Hooke energy equals the k -means SSE to 3.9×10^{-16} . On equal-size partitions $E_{\text{tot}} = (n/k)E_{\text{sp}}$ to 2.2×10^{-16} , while on random partitions the ratio $E_{\text{tot}}/E_{\text{sp}}$ ranges over $[29.7, 197.5]$: the normalisation is a genuine choice. The spectrum of $\widetilde{K}$ (per configuration in `results/tables_identity.md`): for the 25 configurations with a potential of negative type it is PSD in 25 (numerically, $\lambda_{\min}/\lambda_{\max} \geq -3.2 \times 10^{-16}$), and for the rest-length potential it is indefinite in 5/5.

E2: same optimiser, same initialisation, same trajectory. *Criterion:* a brute-force spring relaxation that recomputes the total energy from scratch for every candidate move (it sees only springs) and the kernel implementation of Hartigan’s method via (3) (it sees only K), started from the same random labels with the same sweep order, must produce identical label vectors after every sweep and final energies equal to 10^{-9} relative, in all runs. Subsets of 120 points, three initialisations per configuration. *Result:* 90/90 identical trajectories (mean 4.9 sweeps, at most 11), worst final-energy discrepancy 1.4×10^{-14} . In addition, hub-anchored Hooke springs equilibrated alternately (Remark 3.8(b)) reproduced scikit-learn’s Lloyd k -means from the same initial centroids in 15/15 runs, with inertia equal to the spring energy to 4.3×10^{-16} .

potential	configurations reaching E^* (of 5)				restarts reaching E^* (%)			Lloyd f.p.	best
	relax	lloyd+relax	lloyd	naive	relax	lloyd+relax	lloyd	1-move stable	ARI= 1
d^2 (Hooke)	4/5	5/5	3/5	1/5	54	57	31	46/100	4/5
d^2 on the lift ($x, x ^2/s$)	5/5	5/5	4/5	2/5	80	74	69	73/100	5/5
$(d-r)^2$ (rest length)	5/5	4/5	1/5	1/5	53	49	4	4/100	4/5
d (constant tension)	5/5	5/5	2/5	1/5	42	43	30	43/100	5/5
$1 - e^{-d^2/2\ell^2}$ (saturating)	4/5	2/5	2/5	0/5	4	2	2	12/100	2/5
$\log(1 + d^2/\ell^2)$ (soft)	5/5	4/5	2/5	1/5	37	34	17	26/100	4/5

Table 1: E3 by potential (five datasets each). “Reaching E^* ” means within 10^{-6} relative of the best energy found by any method. “Lloyd f.p. 1-move stable”: Lloyd fixed points that no single move improves (all relaxation fixed points are Lloyd fixed points, Proposition 3.9(ii); 100/100 per potential). “best ARI= 1”: configurations in which the best partitions of relax and lloyd+relax coincide.

E3: relaxation versus kernel k -means on the same energy. Five optimisers see the same kernel: *lloyd* (batch Lloyd on $\widetilde{K}$, minimally shifted only when indefinite, 5 configurations), *lloyd+relax* (Lloyd, then single-move polish), *relax* (single-particle relaxation from the initial labels), *anneal* (Metropolis on the spring energy, 40 sweeps of geometric cooling from the median $|\Delta E|$ of random moves down by 10^{-3} , then relaxation [17]; Hofmann and Buhmann [15] anneal the same energy deterministically) and the control *lloyd (naive)* (Lloyd on $-A + \sigma I$, Proposition 3.10). Initialisations are shared: 10 random labelings and 10 kernel k -means++ seedings [1] per configuration; annealing uses the first 5 random ones. E^* is the best energy found by any method in a configuration. *Criteria (predefined)*: (C3a) every relaxation and annealing fixed point is Voronoi-stable for the kernel used (Proposition 3.9 predicts 100%); (C3b) the relaxation is called distinguishable from the kernel family only if its best-of-restarts energy differs from that of lloyd+relax by more than 10^{-6} relative, either sign being reported since it would be a statement about optimisers of one function; (C3c) the fraction of Lloyd fixed points that are not single-move stable is reported. *Disclosure*: a pilot run used $-A + \sigma I$ as the only Lloyd kernel and found it inert (Proposition 3.10); $\widetilde{K}$ then replaced it as the baseline, which favours the kernel side of the comparison, and the naive shift was kept as a control. C3a’s wording was updated to name the kernel used; its prediction (100%, valid for every PSD representative by Proposition 3.9(ii)), C3b, C3c and all thresholds were written before the pilot and not changed; every number reported comes from the final code. *Results* (Table 1; per configuration in `results/tables_relaxation.md`, and the relative excess energies of the four Lloyd and relaxation optimisers per dataset and potential in `figures/energies.pdf`): all 600/600 relaxation and 150/150 annealing fixed points are Voronoi-stable (C3a). Under C3b, 6 of 30 configurations are distinguishable (5 in favour of the relaxation, 1 against; largest relative difference 7.4×10^{-3} ; the split is optimiser noise: the `-fast` run gives 3–3 with the same 6/30); whenever the two best energies agree the best partitions coincide (ARI = 1; 0 exceptions), and overall ARI = 1 in 24/30. The relaxation reaches E^* in 28 configurations, lloyd+relax in 25, Lloyd alone in 14 and the naive shift in 6 (annealing, with 5 restarts only, is not comparable on this count: 20). Only 34% of Lloyd fixed points on the canonical kernel are single-move stable, 6% on the naive shift (C3c); median excesses and the measured single-move stability of the single-move methods are in `results/tables_relaxation.md`. The best energies were recomputed spring by spring, worst discrepancy 4.3×10^{-13} . E3 is descriptive (counts over 30 configurations, modest restart budgets); no inferential claim is made.

Reading of E1–E3 for the research line. If the screened formulations were per-particle energies of the pairwise kind of Definition 2.1, the screening could not have found a mechanical effect: the spring energies *are* the kernel objectives (E1), the mechanical implementation of the relaxation *is* a kernel implementation (E2), and every relaxed state *is* a kernel k -means fixed point (E3). Differences between optimisers of the same function are visible (Lloyd alone

claim	status
J is invariant under diagonal shift (up to $\sigma(n-k)$), constant and separable terms (Lemma 3.1)	elementary; cf. 8, 23
Lemma 3.1 lists all value invariances of J for $2 \leq k \leq n-1$; all representatives and the minimal PSD shift (Prop. 3.2, Cor. 3.3)	proved here (elementary); not yet refereed; converse of 23, not found in the literature we know
Per-particle spring energy = pairwise-clustering cost $= \frac{1}{2}J_{-A} + \frac{1}{2}(n-k)\phi(0)$, PSD after a shift (Thm. 3.4(b))	literature (15; constant-shift embedding 23; ratio association $\leftrightarrow$ kernel k -means 7, 8); constant made explicit
Zero-rest-length Hooke energy = k -means SSE (Thm. 3.4(c))	classical
Total energy / released tension = min-sum k -clustering / max- k -cut (Thm. 3.4(d))	elementary
Every kernel k -means objective is a Hooke energy of a lifted configuration (Thm. 3.4(e))	classical
Negative-type springs: dataset-independent PD kernel; Hooke springs after a lift and a $1/\sqrt{2}$ rescaling (Thm. 3.4(f))	literature (2, 10, 29); written for spring potentials
d^2 is of negative type; Bernstein transforms preserve negative type (Prop. 3.5(i)–(iii))	classical / literature
$(d-r)^2$ is not of negative type (Prop. 3.5); it has no dataset-independent PD representative, even up to a data-dependent factor (Cor. 3.6); same-minimiser version (Remark 3.7)	proved here (elementary); Cor. 3.6 and Remark 3.7(a) not yet refereed; Remark 3.7(c) open
Lifting only changes the kernel; anchored Hooke springs are Lloyd k -means (Remark 3.8)	trivial / classical
Single-particle relaxation is Hartigan’s method in kernel space (Prop. 3.9(i))	literature (kernel k -groups, 10)
Its fixed points are Voronoi-stable for every PSD representative (Prop. 3.9(ii))	proved here (kernel form of 31; priority not checked)
Lloyd’s rule depends on the PSD representative; large shifts freeze it (Prop. 3.10)	proved here; effect noted by Dhillon et al. [7, 8]
The per-particle triangle-area three-body term and cluster-mean rest length, and the per-spring average, are not w -pairwise for $w \in \{1, 1/n_c\}$; the two normalisations differ (Thm. 4.1; values, not minimisers)	proved here (exact arithmetic, two explicit configurations)
$\sum_c w(n_c) \sum A_{ij}$ is weighted kernel k -means for all A with fixed point weights iff $w \propto 1/m$ ($3 \leq k \leq n-2$; $k=2$ partly open; Prop. 4.2); E_{tot} with $A \geq 0$ is not, even with data-dependent weights (Prop. 4.3; values, not minimisers)	proved here; not yet refereed
Dynamic mechanical clustering with unique minimisers (Problem 4.4); no single configuration with distinct distances is a counterexample (Remark 4.5)	open, not tested; remark proved here, not yet refereed
Any statement about clustering quality on real data	not addressed

Table 2: Mathematical claims and their status. “Not yet refereed”: proved here in v0.4 and checked by exact rank computations for small n (`theory/check_invariances.py` and an independent script), but not yet by an independent referee.

reaches E^* less often than the single-move methods, consistent with França et al. [10], who found Hartigan’s method preferable to Lloyd’s for kernel k -means; the naive shift is worse still), and they are the only differences there are. For the total normalisation the identity is with min-sum k -clustering, which is not a weighted kernel k -means objective (Proposition 4.3); a kernel control on the same matrix then optimises a different function, and a null result against it would be an empirical finding, not a corollary.

6 What can be claimed

Table 2 collects the mathematical claims with their status; it separates what is known, what is proved here and what was only measured (Section 5; “verified” there means reproduced by the scripts of Appendix A with fixed seeds, not proved). Most of what closes the screening is literature.

Limitations. The family of Section 2 is a reconstruction from the line’s public summary; if the original screening used many-body terms, assignment-dependent parameters or a dynamics on

positions, Section 4 applies and the theorem does not. The theorem is about objectives; it does not say that all optimisers of a spring energy behave alike, and E3 shows that they do not. The normalisation family is $\{1, 1/n_c\}$ (by Proposition 4.2, $1/n_c$ is the only size weight inside weighted kernel k -means with fixed point weights; $k = 2$ and data-dependent weights for decreasing $w \neq 1/m$ remain open). The results on invariances and size weights (Propositions 3.2, 4.2, 4.3, Corollaries 3.3, 3.6, Remarks 3.7, 4.5) are new and not yet checked by an independent referee; the relation of σ_* to the shift of Roth et al. [23], and whether the converse is in the literature, were not checked against full texts. Theorem 4.1 certifies three functionals on two configurations and concerns objective values, not minimisers; it says nothing about classes of terms or about how far from pairwise those energies are on typical data. The Bernstein composition result is cited, not re-proved in full. The attributions to França et al. [10] and to the diagonal-shift discussion of Dhillon et al. [7, 8] were checked against abstracts and bibliographic indices, not against the full texts. E3 uses one implementation of each optimiser with modest restart budgets; its numbers describe optimiser luck on five datasets, not a benchmark. Nothing is claimed about the dynamic route beyond its formulation.

Next steps for the research line. (a) Confirm with the author of the line that the screened formulations were of the pairwise form of Definition 2.1, and with which normalisation; if not pairwise, the emphasis moves to Section 4. (b) Settle Open problem 4.4 for the Gaussian flow, by configurations with repeated distances on which no fixed pair function makes $P_{T,\varepsilon}$ the unique minimiser (a Farkas certificate, Remark 4.5), or by a representation. (c) The case $k = 2$ of Proposition 4.2 when $h(a) + h(n - a)$ is constant, and data-dependent weights for decreasing w other than $1/m$. (d) Whether the rest-length spring has a dataset-independent PD kernel with the same minimisers (Remark 3.7). (e) Check against the text of Roth et al. [23] the relation of their minimal shift to σ_* , and whether Proposition 3.2 is stated there or elsewhere. (f) If (a) is confirmed, update the line’s summary to “no mechanical component of the pairwise, per-particle kind can survive”.

A Reproducibility

`experiments/mechanical.py` implements the potentials, energies, kernel forms and optimisers; `experiments/identity_check.py` runs E1, E2 and E4 and `experiments/relaxation.py` runs E3 (both take `-fast`); `experiments/make_numbers.py` converts `results/*.json` into the macros and table bodies used here, so every number in the text is generated, and `results/*.md` hold the full tables. Seeds 20260930 (E1, E2), 20260931 (E3) and 20260932 (E4, own generator); Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 [32], scikit-learn 1.9.1 [20]; 159 s (E1, E2, E4) and 128 s (E3) on shared cores. Implementation details (thresholds, tolerances, the E4 generator and exact elimination over $\mathbb{Q}$) are in the README.

B Proofs of the results on invariances and size weights

Proof of Proposition 3.2. If. Lemma 3.1, with value $\sigma(n - k)$ (the matrix $a\mathbf{1}\mathbf{1}^\top$ is the case $g = \frac{a}{2}\mathbf{1}$). *Uniqueness.* If $\sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top = 0$, the off-diagonal entries give $g_i + g_j = 0$ for $i \neq j$; with a third index l , $g_i = \frac{1}{2}[(g_i + g_j) + (g_i + g_l) - (g_j + g_l)] = 0$, and then $\sigma = 0$. *Only if.* Put $f = \frac{1}{2}(\Delta_{11}, \dots, \Delta_{nn})$ and $e = \Delta - f\mathbf{1}^\top - \mathbf{1}f^\top$, which has zero diagonal and, by Lemma 3.1, $J_e = J_\Delta$. For zero diagonal, $J_e(C) = -2\mathcal{E}(C)$ with $\mathcal{E}(C) = \sum_c \frac{1}{n_c} \sum_{i < j \in C_c} e_{ij}$, so $\mathcal{E}$ is constant on k -partitions. We show $e_{pi} = e_{pj}$ for all distinct p, i, j . If $k \geq 3$, the set $R = [n] \setminus \{p, i, j\}$ has $n - 3 \geq k - 2 \geq 1$ elements; fix a partition of R into $k - 2$ non-empty blocks and compare the partitions $\{\{p, i\}, \{j\}, R\text{-blocks}\}$ and $\{\{p, j\}, \{i\}, R\text{-blocks}\}$: their energies differ by $\frac{1}{2}(e_{pi} - e_{pj})$, which must vanish. If $k = 2$, let $U = [n] \setminus \{i, j\} \neq \emptyset$ and $\gamma_x = e_{ix} - e_{jx}$ for $x \in U$. For $P \subseteq U$ and $Q = U \setminus P$, the partitions $\{P \cup \{i\}, Q \cup \{j\}\}$ and $\{P \cup \{j\}, Q \cup \{i\}\}$ have energies

differing by $\frac{1}{|P|+1} \sum_{x \in P} \gamma_x - \frac{1}{|Q|+1} \sum_{x \in Q} \gamma_x$. With $P = \emptyset$ this gives $\sum_{x \in U} \gamma_x = 0$; with $P = \{p\}$ it gives $\gamma_p(\frac{1}{2} + \frac{1}{n-2}) = 0$, so $e_{ip} = e_{jp}$. In both cases, any two pairs are linked by pairs sharing an index ($e_{ab} = e_{ad} = e_{cd}$), so $e_{ij} = -\sigma$ for all $i \neq j$ and some σ , i.e. $e = \sigma I - \sigma \mathbf{1}\mathbf{1}^\top$ and $\Delta = \sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top$ with $g = f - \frac{\sigma}{2}\mathbf{1}$. All k . The condition for all partitions implies the one for $k = 2$, hence the form; conversely that form gives $\sigma(n-k)$ for every k . *Equivalent forms.* D is linear with kernel $\{g\mathbf{1}^\top + \mathbf{1}g^\top\}$ (if $D(K) = 0$ then $K_{ij} = \frac{1}{2}(K_{ii} + K_{jj})$), and $D(\sigma I) = 2\sigma(\mathbf{1}\mathbf{1}^\top - I)$; likewise $M \mapsto PMP$ has the same kernel on symmetric matrices and $P(\sigma I)P = \sigma P$. $\square$

Proof of Corollary 3.6(c). Now $\kappa[X] = \lambda_X(-A[X]) + \sigma_X I + g_X \mathbf{1}^\top + \mathbf{1}g_X^\top$. For $X \subseteq Y$, restriction gives $(\lambda_X - \lambda_Y)A[X] \in V_{|X|,2}$. The matrix $A[X_s]$ is not in $V_{4,2}$, because $D(A[X_s])_{ij} = 2\phi(0) - 2\phi(d_{ij})$ is not constant off the diagonal ($(s-r)^2, (2s-r)^2, (3s-r)^2$ are never all equal for $s > 0$). Hence $\lambda_{X_s} = \lambda_{X_s \cup X_{s'}} = \lambda_{X_{s'}} =: \lambda$ for all $s, s' > 0$, the same restriction argument gives one σ for all X_s , and κ/λ fails (b) on X_s for large s . $\square$

Proof of Proposition 4.2. (iii) $\Rightarrow$ (i): with $w(m) = \lambda/m$, $F_A^w = \lambda E_{\text{sp}}$ for the zero-diagonal A , and $E_{\text{sp}} = \frac{1}{2}J_{-A}$ by Theorem 3.4(b). (i) $\Rightarrow$ (ii) is trivial. (ii) $\Rightarrow$ (i): the matrices $M(x)_{ij} = (x_i - x_j)^2$ satisfy $M(x+y) - M(x) - M(y) = 2[(x_i - x_j)(y_i - y_j)]_{ij}$, which for $x = e_a, y = e_b$ is $-2(E_{ab} + E_{ba})$; so they span all zero-diagonal symmetric matrices, and since M is polynomial, so do the $M(x)$ with distinct coordinates (a non-empty open set). Membership in the linear space $\mathcal{G}_\omega$ passes to linear combinations. (i) $\Rightarrow$ (iii): under (i), the $\binom{n}{2}$ functions $F_{E_{ab}+E_{ba}}^w$ ($a < b$) and the constant lie in $\mathcal{G}_\omega$, of dimension $\leq \binom{n}{2}$, so they are linearly dependent: some $e \neq 0$ (zero diagonal, symmetric) has F_e^w constant. The exchange of the proof of Proposition 3.2, with energies weighted by w , gives $w(2)(e_{pi} - e_{pj}) = 0$, so $e \equiv \varepsilon \neq 0$ off the diagonal and $\sum_c h(n_c)$ is constant over all sizes $(n_1, \dots, n_k)$ of k -partitions. Compare the size vectors $(m+1, 1, \rho)$ and $(m, 2, \rho)$ with the same $k-2 \geq 1$ remaining sizes ρ (possible for $1 \leq m \leq n-k$): $h(m+1) - h(m) = h(2) - h(1) = h(2)$, so $h(m) = (m-1)h(2)$, i.e. $w(m) = 2w(2)/m$. For $k = 2$ the exchange with $P = \emptyset$ and $P = \{x\}$ gives $w(n-1)\sum_U \gamma = 0$ and $(w(2) + w(n-2))\gamma_x = 0$, hence again $e \equiv \varepsilon$, and $\sum_c h(n_c) = h(a) + h(n-a)$; for $n = 4$, $h(1) + h(3) = 2h(2)$ reads $3w(3) = 2w(2)$. $\square$

Proof of Proposition 4.3. Take $a \neq b$ with $A_{ab} > 0$. Split $[n]$ into $k+2$ non-empty groups with a, b in different groups; keep $k-2$ groups $R_1, \dots, R_{k-2}$ as fixed clusters and call the other four $G_1, \dots, G_4$, with $a \in G_1, b \in G_2$; for the second statement use singletons $G_1 = \{a\}, G_2 = \{b\}, G_3 = \{c\}, G_4 = \{d\}$ (possible exactly when $k \geq 3$ or $n = 4$). For each of the seven partitions $\{X, Y\}$ of $\{1, 2, 3, 4\}$ into two non-empty sets, let C_X be the k -partition with clusters $G_X = \bigcup_{t \in X} G_t$, G_Y and the R 's, and put $\nu(C_X) = \varepsilon(X)\hat{\omega}(X)\hat{\omega}(Y)$, where $\hat{\omega}(X) = \omega(G_X)$, $\varepsilon(X) = +1$ if $|X| \in \{1, 3\}$ and $\varepsilon(X) = -1$ if $|X| = 2$; $\nu = 0$ on all other partitions. (1) $\sum_X \nu(C_X) = (\hat{\Omega}^2 - \sum_t \hat{\omega}_t^2) - 2\sum_{t < u} \hat{\omega}_t \hat{\omega}_u = 0$, with $\hat{\Omega} = \sum_t \hat{\omega}_t$, because each pair $t < u$ is separated by exactly two of the three 2+2 splits. (2) ν annihilates $\mathcal{G}_\omega$: by (1) the terms of $J_K^\omega(C_X)$ that are common to the seven partitions drop out, and what remains is $-\sum_{S \in \{X, Y\}} \hat{Q}(S)/\hat{\omega}(S)$ with $\hat{Q}(S) = \sum_{t, u \in S} \hat{B}_{tu}$, $\hat{B}_{tu} = \sum_{i \in G_t, j \in G_u} \omega_i \omega_j K_{ij}$. Every non-empty proper $S \subset \{1, 2, 3, 4\}$ is a block of exactly one C_X , and $\nu(C_X)/\hat{\omega}(S) = \varepsilon(S)\hat{\omega}(S^c)$; so it suffices that $\sum_{S \ni t, u} \varepsilon(S)\hat{\omega}(S^c) = 0$ for all t, u . For $t = u$: $(\hat{\Omega} - \hat{\omega}_t) - \sum_{v \neq t} (\hat{\Omega} - \hat{\omega}_t - \hat{\omega}_v) + \sum_{v \neq t} \hat{\omega}_v = 0$; for $t \neq u$: $-\hat{\omega}(\{t, u\}^c) + \hat{\omega}(\{t, u\}^c) = 0$. (3) For E_{tot} , $E_{\text{tot}}(C_X) = \text{const} + \sum_{t < u, t, u \text{ on the same side}} \hat{A}_{tu}$ with $\hat{A}_{tu} = \sum_{i \in G_t, j \in G_u} A_{ij} \geq 0$, and $\hat{A}_{12} \geq A_{ab} > 0$. The pair $\{t, u\}$ lies in a two-group block once $(-\hat{\omega}(\{t, u\})\hat{\omega}(\{v, z\}))$ and in a three-group block twice $(\hat{\omega}_v(\hat{\Omega} - \hat{\omega}_v) + \hat{\omega}_z(\hat{\Omega} - \hat{\omega}_z))$, where $\{v, z\}$ is the complementary pair; hence $\langle \nu, E_{\text{tot}} \rangle = 2\sum_{t < u} \hat{A}_{tu} \hat{\omega}_v \hat{\omega}_z \geq 2A_{ab} \hat{\omega}_3 \hat{\omega}_4 > 0$. Since $\langle \nu, J_K^\omega + \text{const} \rangle = 0$, E_{tot} is not of that form. With singleton groups and a size weight, the same count gives $\langle \nu, F_A^w \rangle = \sum_{t < u} A_{tu} [(w(3) - w(2))\hat{\omega}(\{t, u\})\hat{\omega}(\{v, z\}) + 2w(3)\hat{\omega}_v \hat{\omega}_z] > 0$ under the stated conditions on w . $\square$

The annihilation and the pairing formula were checked in exact arithmetic for $n = 4$ with singleton groups (`theory/check_invariances.py`) and for $n = 7, k = 3$ with non-singleton

groups (the integrator’s script, see the README).

References

- [1] D. Arthur and S. Vassilvitskii. k -means++: the advantages of careful seeding. In *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1027–1035, 2007.
- [2] C. Berg, J. P. R. Christensen, and P. Ressel. *Harmonic Analysis on Semigroups: Theory of Positive Definite and Related Functions*, volume 100 of *Graduate Texts in Mathematics*. Springer, New York, 1984.
- [3] M. Blatt, S. Wiseman, and E. Domany. Superparamagnetic clustering of data. *Physical Review Letters*, 76(18):3251–3254, 1996.
- [4] Y. Cheng. Mean shift, mode seeking, and clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(8):790–799, 1995.
- [5] D. Comaniciu and P. Meer. Mean shift: a robust approach toward feature space analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(5):603–619, 2002.
- [6] I. S. Dhillon, Y. Guan, and B. Kulis. Kernel k -means: spectral clustering and normalized cuts. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD ’04)*, pages 551–556. ACM, 2004.
- [7] I. S. Dhillon, Y. Guan, and B. Kulis. A unified view of kernel k -means, spectral clustering and graph cuts. UTCS Technical Report TR-04-25, Department of Computer Sciences, University of Texas at Austin, 2005. Dated 18 February 2005.
- [8] I. S. Dhillon, Y. Guan, and B. Kulis. Weighted graph cuts without eigenvectors: a multilevel approach. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 29(11):1944–1957, 2007.
- [9] R. Durbin and D. Willshaw. An analogue approach to the travelling salesman problem using an elastic net method. *Nature*, 326:689–691, 1987.
- [10] G. França, M. L. Rizzo, and J. T. Vogelstein. Kernel k -groups via Hartigan’s method. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(12):4411–4425, 2021. doi: 10.1109/TPAMI.2020.2998120.
- [11] K. Fukunaga and L. D. Hostetler. The estimation of the gradient of a density function, with applications in pattern recognition. *IEEE Transactions on Information Theory*, 21(1):32–40, 1975.
- [12] M. Girolami. Mercer kernel-based clustering in feature space. *IEEE Transactions on Neural Networks*, 13(3):780–784, 2002.
- [13] J. A. Hartigan. *Clustering Algorithms*. Wiley, New York, 1975.
- [14] J. A. Hartigan and M. A. Wong. Algorithm AS 136: A k -means clustering algorithm. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 28(1):100–108, 1979.
- [15] T. Hofmann and J. M. Buhmann. Pairwise data clustering by deterministic annealing. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 19(1):1–14, 1997.
- [16] D. Horn and A. Gottlieb. Algorithm for data clustering in pattern recognition problems based on quantum mechanics. *Physical Review Letters*, 88(1):018702, 2002.
- [17] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi. Optimization by simulated annealing. *Science*, 220(4598):671–680, 1983.
- [18] S. P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [19] J. MacQueen. Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 281–297. University of California Press, 1967.
- [20] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [21] K. Rose. Deterministic annealing for clustering, compression, classification, regression, and related optimization problems. *Proceedings of the IEEE*, 86(11):2210–2239, 1998.
- [22] K. Rose, E. Gurewitz, and G. C. Fox. Statistical mechanics and phase transitions in clustering. *Physical Review Letters*, 65(8):945–948, 1990.
- [23] V. Roth, J. Laub, M. Kawanabe, and J. M. Buhmann. Optimal cluster preserving embedding of nonmetric proximity data. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 25(12):1540–1551, 2003.
- [24] S. Sahni and T. Gonzalez. P-complete approximation problems. *Journal of the ACM*, 23(3):555–565, 1976.
- [25] R. L. Schilling, R. Song, and Z. Vondráček. *Bernstein Functions: Theory and Applications*, volume 37 of *De Gruyter Studies in Mathematics*. De Gruyter, Berlin, 2nd edition, 2012.
- [26] I. J. Schoenberg. Remarks to Maurice Fréchet’s article “sur la définition axiomatique d’une classe d’espaces distanciés vectoriellement applicable sur l’espace de Hilbert”. *Annals of Mathematics*, 36(3):724–732, 1935.
- [27] I. J. Schoenberg. Metric spaces and positive definite functions. *Transactions of the American Mathematical Society*, 44(3):522–536, 1938.
- [28] B. Schölkopf, A. Smola, and K.-R. Müller. Non-linear component analysis as a kernel eigenvalue problem. *Neural Computation*, 10(5):1299–1319, 1998.

- [29] D. Sejdinovic, B. Sriperumbudur, A. Gretton, and K. Fukumizu. Equivalence of distance-based and RKHS-based statistics in hypothesis testing. *The Annals of Statistics*, 41(5):2263–2291, 2013. doi: 10.1214/13-AOS1140.
- [30] S. Z. Selim and M. A. Ismail. K-means-type algorithms: a generalized convergence theorem and characterization of local optimality. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 6(1):81–87, 1984.
- [31] M. Telgarsky and A. Vattani. Hartigan’s method: k -means clustering without Voronoi. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 9 of *JMLR Workshop and Conference Proceedings*, pages 820–827, 2010.
- [32] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, I. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17: 261–272, 2020.
- [33] W. E. Wright. Gravitational clustering. *Pattern Recognition*, 9(3):151–166, 1977.
- [34] A. L. Yuille. Generalized deformable models, statistical physics, and matching problems. *Neural Computation*, 2(1):1–24, 1990.
- [35] H. Zha, X. He, C. Ding, M. Gu, and H. D. Simon. Spectral relaxation for k -means clustering. In *Advances in Neural Information Processing Systems 14 (NIPS 2001)*, pages 1057–1064. MIT Press, 2002.